

Supplementary Material: Operator-Space Dynamics of Financial Markets

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April 7, 2026

This Supplementary Material provides: (S1) Discussion about exponent alpha (S2) connection between GLE memory kernels and diffusion exponents; (S3) adiabatic eigenvector rotation near spectral crossings; (S4) window-overlap bias analysis; (S5) individual asset results for all 13 assets; (S6) robustness checks; and (S7) proposed additional figures and tables.

I Mori–Zwanzig Formalism and Matrix FDT

Let $P(t)$ be the log-returns centered by the local mean $\mu(\tau)$ within a sliding window. The resulting Hankel covariance operator $\mathbf{K}(\tau) \in \mathbb{R}^{L \times L}$ is, by construction, a fluctuation operator. Its evolution is governed by the Liouvillian $\mathcal{L}$, such that

$$\frac{d}{d\tau} \mathbf{K}(\tau) = i\mathcal{L}\mathbf{K}(\tau), \quad (1)$$

with formal solution $\mathbf{K}(\tau) = e^{i\mathcal{L}\tau} \mathbf{K}(0)$.

To extract the effective dynamics of $\mathbf{K}(\tau)$, we introduce a projection operator $\mathcal{P}$ acting on the Liouville space of matrices, defined via the Frobenius inner product $\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr}(\mathbf{A}^\dagger \mathbf{B})$:

$$\mathcal{P}\mathbf{B} = \frac{\text{Tr}(\mathbf{B}^\dagger \mathbf{K}(0))}{\text{Tr}(\mathbf{K}(0)^\dagger \mathbf{K}(0))} \mathbf{K}(0). \quad (2)$$

We also define the complementary projector $\mathcal{Q} = \mathcal{I} - \mathcal{P}$. This induces a decomposition of the Liouvillian dynamics into resolved and orthogonal components.

Applying $\mathcal{P}$ and $\mathcal{Q}$ to the evolution equation yields the coupled system

$$\frac{d}{d\tau} \mathcal{P}\mathbf{K}(\tau) = \mathcal{P}i\mathcal{L}\mathcal{P}\mathbf{K}(\tau) + \mathcal{P}i\mathcal{L}\mathcal{Q}\mathbf{K}(\tau), \quad (3)$$

$$\frac{d}{d\tau} \mathcal{Q}\mathbf{K}(\tau) = \mathcal{Q}i\mathcal{L}\mathcal{P}\mathbf{K}(\tau) + \mathcal{Q}i\mathcal{L}\mathcal{Q}\mathbf{K}(\tau). \quad (4)$$

The orthogonal component admits the formal solution

$$\mathcal{Q}\mathbf{K}(\tau) = e^{i\mathcal{Q}\mathcal{L}\tau} \mathcal{Q}\mathbf{K}(0) + \int_0^\tau e^{i\mathcal{Q}\mathcal{L}(\tau-s)} \mathcal{Q}i\mathcal{L}\mathcal{P}\mathbf{K}(s) ds. \quad (5)$$

Substituting this expression back into the projected equation leads to the exact Mori–Zwanzig [1, 2, 3] identity for $\mathbf{K}(\tau)$:

$$\frac{d}{d\tau} \mathbf{K}(\tau) = \mathcal{P}i\mathcal{L}\mathbf{K}(\tau) + \int_0^\tau \mathcal{P}i\mathcal{L}e^{i\mathcal{Q}\mathcal{L}(\tau-s)} \mathcal{Q}i\mathcal{L}\mathbf{K}(s) ds + \boldsymbol{\xi}(\tau), \quad (6)$$

where the fluctuating force is defined as

$$\boldsymbol{\xi}(\tau) = \mathcal{P}i\mathcal{L}e^{i\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}\mathbf{K}(0). \quad (7)$$

By construction, the random force is orthogonal to the resolved variable,

$$\text{Tr}(\boldsymbol{\xi}(\tau)^\dagger \mathbf{K}(0)) = 0, \quad (8)$$

ensuring consistency of the projection.

The memory kernel naturally emerges as

$$\boldsymbol{\Gamma}(\tau) = -\mathcal{P}i\mathcal{L}e^{i\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}i\mathcal{L}\mathcal{P}. \quad (9)$$

Since the projector $\mathcal{P}$ is rank-one in the space spanned by $\mathbf{K}(0)$, the dynamics reduces to a convolution acting on $\mathbf{K}(\tau)$, yielding the matrix-valued Generalized Langevin Equation (GLE):

$$\frac{d}{d\tau}\mathbf{K}(\tau) = -\int_0^\tau \boldsymbol{\Gamma}(\tau-s)\mathbf{K}(s)ds + \boldsymbol{\xi}(\tau) \quad (10)$$

Finally, the consistency of the reduced dynamics imposes a matrix-valued fluctuation–dissipation relation,

$$\boldsymbol{\Gamma}(\tau) = \langle \boldsymbol{\xi}(\tau)\boldsymbol{\xi}(0)^\dagger \rangle \cdot \langle \mathbf{K}(0)\mathbf{K}(0)^\dagger \rangle^{-1}, \quad (11)$$

with

$$\langle \mathbf{K}(0)\mathbf{K}(0)^\dagger \rangle = \|\mathbf{K}\|_F^2. \quad (12)$$

This establishes the exact correspondence between dissipation and fluctuations at the operator level.

II Matrix Derivation of Diffusion Exponents

To derive the scaling of the cumulative operator

$$\mathbf{X}(\tau) = \int_0^\tau \mathbf{K}(s) ds, \quad (13)$$

we analyze its variance in the Frobenius manifold,

$$\Sigma^2(\tau) = \mathbb{E}[\|\mathbf{X}(\tau)\|_F^2]. \quad (14)$$

Expanding the norm explicitly,

$$\|\mathbf{X}(\tau)\|_F^2 = \int_0^\tau \int_0^\tau \text{Tr}(\mathbf{K}(s)^\dagger \mathbf{K}(s')) ds ds'. \quad (15)$$

Assuming stationarity of the fluctuations, the two-time correlation depends only on the lag, and we define the matrix correlation function

$$C(\tau) = \text{Tr} \langle \mathbf{K}(\tau)^\dagger \mathbf{K}(0) \rangle. \quad (16)$$

This reduces the variance to the convolution form

$$\Sigma^2(\tau) = 2 \int_0^\tau (\tau-s) C(s) ds. \quad (17)$$

We now derive the evolution equation for $C(\tau)$ from the matrix-valued Generalized Langevin Equation Eq.(10), multiplying by $\mathbf{K}(0)$, taking the trace and ensemble average, and using the orthogonality condition $\text{Tr}(\boldsymbol{\xi}(\tau)^\dagger \mathbf{K}(0)) = 0$, we obtain

$$\frac{d}{d\tau}C(\tau) = - \int_0^\tau \text{Tr}\langle \boldsymbol{\Gamma}(\tau-s) \mathbf{K}(s) \mathbf{K}(0)^\dagger \rangle ds. \quad (18)$$

Since the projection operator is rank-one in the space spanned by $\mathbf{K}(0)$, the dynamics effectively reduces to a scalar convolution, and we define the effective kernel

$$\gamma(\tau) = \frac{1}{L} \text{Tr}(\boldsymbol{\Gamma}(\tau)), \quad (19)$$

yielding the closed Volterra equation

$$\dot{C}(\tau) = - \int_0^\tau \gamma(s) C(\tau-s) ds. \quad (20)$$

The asymptotic behaviour of $C(\tau)$ is governed by the long-time properties of the memory kernel. If the kernel exhibits power-law decay,

$$\gamma(\tau) \sim \tau^{-(2-\nu)}, \quad 0 < \nu < 1, \quad (21)$$

then standard results for non-Markovian GLEs imply

$$C(\tau) \sim \tau^{-\nu}. \quad (22)$$

Substituting into the variance expression,

$$\Sigma^2(\tau) = 2 \int_0^\tau (\tau-s) s^{-\nu} ds \sim \tau^{1+\nu}, \quad (23)$$

which yields the diffusion exponent

$$\beta = 1 + \nu. \quad (24)$$

This result can be understood more generally by recalling the classification of diffusion regimes for scalar GLEs [4]. For a process

$$\dot{x}(t) = - \int_0^t \Gamma(t-t') x(t') dt' + \xi(t), \quad (25)$$

satisfying the fluctuation-dissipation relation $\mathbb{E}[\xi(t)\xi(0)] = k_B T \Gamma(t)$, the long-time scaling $\mathbb{E}[x^2(t)] \sim t^\beta$ is determined by the low-frequency behaviour of the memory kernel $\hat{\Gamma}(0) = \int_0^\infty \Gamma(t) dt$:

- If $\hat{\Gamma}(0) < \infty$, the effective friction is finite and $\beta = 1$ (normal diffusion).
- If $\hat{\Gamma}(0) = \infty$ and $\Gamma(t) \sim t^{-(2-\nu)}$ with $0 < \nu < 1$, then $\beta = 1 + \nu > 1$ (superdiffusion).
- If $\hat{\Gamma}(0) = 0$, subdiffusive regimes with $\beta < 1$ may arise.

In the matrix GLE, these regimes apply to each dynamical block. The structural component $\boldsymbol{\Gamma}_s(\tau)$ satisfies $\hat{\Gamma}_s(0) < \infty$, leading to $\beta_s = 1$, whereas the bulk component exhibits long-memory behaviour $\boldsymbol{\Gamma}_b(\tau) \sim \tau^{-(2-\nu)}$, yielding superdiffusive exponents $\beta_b \in [1.3, 1.7]$.

In practice, the short-time transient regime can bias the estimation of β . Following standard procedures, we discard the lower 30% of lags and perform log-log regression on the upper 70%, where the mean-square displacement has reached its asymptotic scaling regime.

III Adiabatic Eigenvector Dynamics

We analyze the slow-time evolution of the eigenvectors of the fluctuation operator $\mathbf{K}(\tau)$. Let $(\lambda_k(\tau), \varphi_k(\tau))$ denote the eigenpairs,

$$\mathbf{K}(\tau)\varphi_k(\tau) = \lambda_k(\tau)\varphi_k(\tau), \quad (26)$$

where τ is the slow time indexing the sliding window, distinct from the internal time used to construct $\mathbf{K}$. We assume that $\mathbf{K}(\tau)$ is differentiable in τ , corresponding to a slowly varying regime.

Differentiating the eigenvalue equation with respect to τ yields

$$\frac{\partial \mathbf{K}}{\partial \tau} \varphi_k + \mathbf{K} \frac{\partial \varphi_k}{\partial \tau} = \frac{d\lambda_k}{d\tau} \varphi_k + \lambda_k \frac{\partial \varphi_k}{\partial \tau}. \quad (27)$$

Projecting onto another eigenvector φ_j and using the orthonormality $\langle \varphi_j, \varphi_k \rangle = \delta_{jk}$, we obtain

$$\langle \varphi_j, \partial_\tau \mathbf{K} \varphi_k \rangle + \lambda_j \langle \varphi_j, \partial_\tau \varphi_k \rangle = \lambda_k \langle \varphi_j, \partial_\tau \varphi_k \rangle. \quad (28)$$

For $j \neq k$, this gives the off-diagonal coupling

$$\langle \varphi_j, \partial_\tau \varphi_k \rangle = \frac{\langle \varphi_j, \partial_\tau \mathbf{K} \varphi_k \rangle}{\lambda_k - \lambda_j}. \quad (29)$$

Since the eigenvectors form a complete orthonormal basis, we expand

$$\partial_\tau \varphi_k = \sum_j c_{jk} \varphi_j, \quad c_{jk} = \langle \varphi_j, \partial_\tau \varphi_k \rangle. \quad (30)$$

The normalization condition $\langle \varphi_k, \varphi_k \rangle = 1$ implies $c_{kk} = 0$, and therefore

$$\partial_\tau \varphi_k = \sum_{j \neq k} \frac{\langle \varphi_j, \partial_\tau \mathbf{K} \varphi_k \rangle}{\lambda_k - \lambda_j} \varphi_j. \quad (31)$$

This defines the rotation rates

$$R_{jk}(\tau) = \frac{\langle \varphi_j, \partial_\tau \mathbf{K} \varphi_k \rangle}{\lambda_k - \lambda_j}, \quad (32)$$

which control the mixing between eigenmodes. The apparent divergence of R_{jk} as $\lambda_k \rightarrow \lambda_j$ signals a breakdown of adiabatic decoupling and the onset of strong mode mixing.

We now connect this spectral dynamics with the matrix-valued Generalized Langevin Equation,

$$\partial_\tau \mathbf{K}(\tau) = - \int_0^\tau \mathbf{\Gamma}(\tau - s) \mathbf{K}(s) ds + \boldsymbol{\xi}(\tau). \quad (33)$$

Substituting into the expression for R_{jk} yields

$$R_{jk}(\tau) = \frac{\langle \varphi_j, [- \int_0^\tau \mathbf{\Gamma}(\tau - s) \mathbf{K}(s) ds + \boldsymbol{\xi}(\tau)] \varphi_k \rangle}{\lambda_k - \lambda_j}. \quad (34)$$

This decomposition shows that the rotation rates contain both a deterministic memory contribution and a stochastic forcing term. The latter induces angular diffusion in eigenvector space.

The magnitude of R_{jk} is controlled by the spectral gaps $|\lambda_k - \lambda_j|$. In the bulk of the spectrum, eigenvalues exhibit level repulsion and typical spacings scale as $|\lambda_k - \lambda_j| \sim O(L^{-1})$, as expected from random matrix statistics. Consequently,

$$R_{jk}^{(\text{bulk})} \sim O(L), \quad (35)$$

indicating rapid and strongly fluctuating rotations. Since the number of interacting pairs scales as $O(L^2)$, this leads to collective stochastic mixing of bulk eigenvectors.

In contrast, structural modes are spectrally isolated, with $|\lambda_k - \lambda_j| \gg L^{-1}$. Therefore,

$$R_{jk}^{(\text{structural})} = O(1), \quad (36)$$

and the corresponding eigenvectors evolve slowly under the same driving.

This separation of time scales has direct consequences for the memory kernel and diffusion properties. Rapid stochastic rotation in the bulk effectively redistributes correlations across modes, generating long-memory kernels of the form $\Gamma_b(\tau) \sim \tau^{-(2-\nu)}$. As shown in the previous section, this leads to superdiffusive behaviour with exponents $\beta_b \in [1.3, 1.7]$.

Conversely, the slow evolution of isolated structural modes preserves short-range temporal correlations, resulting in kernels with finite integral $\hat{\Gamma}_s(0) < \infty$ and normal diffusion $\beta_s \approx 1$.

Thus, the emergence of distinct diffusion exponents is directly linked to the spectral geometry of $\mathbf{K}(\tau)$ and the resulting hierarchy of eigenvector rotation rates.

IV Optimal Parameters and Intrinsic Market Scales

A Parameter table

Table 1 reports the optimal parameters ($\Delta\tau_{\text{opt}}$, L_{opt}) for all B3 assets analysed, grouped by instrument class. The ratio $\Lambda = L_{\text{opt}}/\Delta\tau_{\text{opt}}$ is also reported; it measures the number of intrinsic relaxation steps contained within one embedding window and provides a dimensionless characterisation of the memory depth of $K(\tau)$ for each asset.

B Discussion

Range and asset-dependence. For liquid assets, $\Delta\tau_{\text{opt}}$ spans [20, 50] candles and L_{opt} spans [30, 150], with means $\bar{\Delta\tau}_{\text{opt}} \approx 34$ and $\bar{L}_{\text{opt}} \approx 89$ candles. These ranges are narrow relative to the full grid explored ($\Delta\tau_0 \in [5, 100]$, $L \in [20, 160]$), suggesting that the parameters are not free but are constrained by the intrinsic autocorrelation structure of $K(\tau)$. The variation that does occur is physically meaningful: assets with faster price dynamics (high-frequency futures such as WIN\$N and WDO\$N) converge to moderate $\Delta\tau_{\text{opt}}$, while assets with slower structural evolution (BDRs tracking foreign equities) tend toward larger L_{opt} .

The ratio $\Lambda = L_{\text{opt}}/\Delta\tau_{\text{opt}}$. The ratio Λ quantifies how many relaxation steps fit within one embedding window. For liquid equities and financial futures, $\Lambda \in [1.2, 5.0]$, with most assets clustering near $\Lambda \approx 2$ –4. This range is consistent with the requirement that the embedding window be long enough to capture the dominant memory of $K(\tau)$ (hence $\Lambda \gg 1$) but short enough to remain within a single dynamical regime (hence Λ not too large). The rate future DI1\$N stands out with $\Lambda = 1.2$ and $L_{\text{opt}} = 30$ — the smallest embedding dimension in the sample — consistent with the interpretation that interest-rate futures have shorter spectral memory than equity or FX instruments, reflecting the mean-reverting nature of rate dynamics.

Table 1: Optimal parameters $\Delta\tau_{\text{opt}}$ (in candles), L_{opt} , and their ratio $\Lambda = L_{\text{opt}}/\Delta\tau_{\text{opt}}$ for all B3 assets studied. Assets are grouped by instrument class and sorted by $\Delta\tau_{\text{opt}}$ within each group. Liquidity tier: H=high, M=moderate, L=low.

Asset	Class	Liq.	$\Delta\tau_{\text{opt}}$	L_{opt}	Λ
<i>Equities</i>					
PETR4	Equity	H	20	80	4.0
ITUB4	Equity	H	20	100	5.0
BBAS3	Equity	H	20	80	4.0
ABEV3	Equity	H	40	60	1.5
VALE3	Equity	H	40	120	3.0
B3SA3	Equity	H	40	110	2.8
BBDC4	Equity	H	50	90	1.8
<i>Financial futures</i>					
BIT\$D	Crypto future	M	20	60	3.0
DI1\$N	Rate future	H	25	30	1.2
WIN\$N	Index future	H	40	70	1.8
WSP\$D	Index future	M	45	60	1.3
WDO\$N	FX future	H	45	80	1.8
T10\$D	Rate (BDR)	M	40	100	2.5
<i>ETFs</i>					
NASD11	Index ETF	M	20	70	3.5
GOLD11	Commodity ETF	M	25	60	2.4
CORN11	Commodity ETF	L	30	90	3.0
BITH11	Crypto ETF	M	40	150	3.8
BBOI11	Commodity ETF	L	60	60	1.0
<i>BDRs</i>					
MELI34	Equity BDR	M	25	120	4.8
NVDC34	Equity BDR	M	25	120	4.8
TSLA34	Equity BDR	M	35	140	4.0
AAPL34	Equity BDR	M	45	60	1.3
BABA34	Equity BDR	L	70	90	1.3
<i>Low-liquidity futures (commodity)</i>					
ICF\$D	Coffee future	L	35	60	1.7
CCM\$D	Corn future	L	35	100	2.9
BGI\$D	Cattle future	L	30	50	1.7
<i>Real State</i>					
MXRF11	Real-estate fund	M	70	60	0.9
SEQR11	Real-estate fund	L	70	80	1.1
<i>Summary (liquid assets only: H+M, excluding FIIs and low-liq. futures)</i>					
Mean	—	—	34	89	2.8
Std	—	—	13	30	1.3

Low-liquidity and FII regimes. Low-liquidity commodity futures (BGI\$D, ICF\$D, CCM\$D) yield parameters comparable to liquid assets in magnitude but with larger estimation uncertainty. FIIs (MXRF11, SEQR11) are qualitatively different: both reach $\Delta\tau_{\text{opt}} = 70$ candles — the largest in the sample — with $\Lambda \approx 1$, indicating that the spectral operator barely resolves a single relaxation cycle within the embedding window at any scale tested. This is consistent with the anomalously large $\hat{\alpha}$ observed for these assets (Section ??): infrequent and uncorrelated regime crossings correspond to a process with very long intrinsic timescales, poorly captured by the M5 candle grid used here.

Timescale invariance. A key property of $(\Delta\tau_{\text{opt}}, L_{\text{opt}})$ is that they are determined from the M5 series (with H1 used only as a cross-validation condition in the score $\mathcal{S}$) yet characterise the operator $K(\tau)$ in a timescale-invariant sense: the same L_{opt} minimises the estimation variance at both M5 and H1 simultaneously, by construction of the criterion. This means that the physical timescale of the embedding window, $T_{\text{embed}} = L_{\text{opt}} \times \delta t_{\text{M5}}$, where $\delta t_{\text{M5}} = 5$ min, can be computed in minutes and compared across assets. For the liquid equity cluster (PETR4, ITUB4, BBAS3), $T_{\text{embed}} \approx 80 \times 5 = 400$ min ≈ 1 trading day; for DI1\$N, $T_{\text{embed}} \approx 30 \times 5 = 150$ min ≈ 2 trading hours. These timescales can be interpreted as the spectral coherence horizon of the market, i.e., the maximum interval over which the eigenstructure of $K(\tau)$ remains approximately invariant. Their magnitude, on the order of days, suggests the presence of an emergent mesoscopic timescale separating microstructural fluctuations from structural regime changes.

V AR(1) Reference Processes and the Exponent α

A Definition and Estimation

At each time step τ , the number of structural eigenvalues is defined as

$$n_s(\tau) = \# \{ \lambda_i(\tau) > \lambda_+ \}, \quad (37)$$

where λ_+ is the upper edge of the Marchenko–Pastur spectral bulk. A *regime-crossing event* occurs whenever $n_s(\tau)$ changes, i.e., whenever at least one eigenvalue crosses λ_+ . Let $\{T_k\}$ denote the sequence of waiting times between successive crossing events. Empirically, this distribution follows a power law over several decades,

$$p(T) \sim T^{-\alpha}, \quad T \gg 1, \quad (38)$$

and the exponent α is estimated by the Hill estimator with automatic plateau detection. Given the ordered waiting times $\tau_{(1)}^* \geq \tau_{(2)}^* \geq \dots \geq \tau_{(n)}^*$, the Hill statistic at threshold order k is

$$\hat{\alpha}(k) = 1 + k \left[\sum_{i=1}^k \ln \frac{\tau_{(i)}^*}{\tau_{(k)}^*} \right]^{-1}, \quad (39)$$

computed for $k = 2, \dots, n-2$ in $\mathcal{O}(n)$ operations via cumulative sums of $\ln \tau^*$. The optimal k is identified as the longest contiguous run of indices over which $|\hat{\alpha}(k+1) - \hat{\alpha}(k)| < \delta$ with $\delta = 0.05$, provided the run length is at least 5. The reported $\hat{\alpha}$ is the mean of $\hat{\alpha}(k)$ over this plateau, and the uncertainty is its standard deviation. When no plateau of sufficient length is found, a fallback estimate is computed using the 75th percentile of τ^* as x_{min} . Confidence intervals should be regarded as indicative; a precise characterisation of the finite-sample bias of the Hill estimator in this setting is left for future work.

B Reference Processes

To calibrate α we construct a family of synthetic reference processes. The AR(1) process

$$x_t = \phi x_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (40)$$

is used to generate the synthetic price series $P_t = \exp(x_t)$, with the same sample length as the empirical series. The parameter $\phi \in [0, 1]$ interpolates continuously between two anchors:

White noise ($\phi = 0$). The series $x_t \sim \mathcal{N}(0, 1)$ i.i.d. represents the complete absence of temporal correlation. It constitutes the *upper anchor*: any genuine temporal structure in the data should lower α below this baseline.

Geometric Brownian Motion ($\phi = 1$). The cumulative sum $x_t = \sum_{s=1}^t \varepsilon_s$ is a random walk, equivalent to the Black–Scholes null model. It constitutes the *lower anchor*: a market indistinguishable from GBM yields α at this level.

C Measured AR(1) Curve

Table 2 reports the measured exponents for the AR(1) family at steps $\phi \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, computed using a synthetic series of the same length as the PETR4 M5 dataset, with fixed parameters $\Delta\tau = 20$ and $L = 80$. The RATIO column reports the energy-balance ratio $R_{\text{FDT}} = -\langle D \rangle / \langle I \rangle$, where $\langle D \rangle$ and $\langle I \rangle$ are the time-averaged dissipation and noise-injection rates of the operator $K(\tau)$, estimated from a sliding window of width $W = 200$ steps over the stored covariance sequence. At thermodynamic equilibrium, $R_{\text{FDT}} = 1$ exactly; deviations signal a net energy flux between the structural and bulk spectral subspaces. For all AR(1) reference processes, $R_{\text{FDT}} = 1.000 \pm 0.001$ to within numerical precision, confirming that the estimator is unbiased on synthetic data and that any departure from unity observed in real assets is a genuine market signature rather than a pipeline artefact. The L_2 column measures the deviation of the empirical eigenvalue density from the Marchenko–Pastur prediction: the large values at small ϕ reflect the accumulation of eigenvalues below λ_- in the Hankel-embedded covariance matrix of a nearly uncorrelated process, a known artefact of Toeplitz structure in the embedding; this artefact diminishes as $\phi \rightarrow 1$ because the embedded matrix of a random walk has a smoother spectral profile.

Table 2: AR(1) reference results. Fixed parameters: $\Delta\tau = 20$, $L = 80$, series length equal to PETR4 M5 dataset. $\hat{\alpha}$: crossing-time exponent estimated by the Hill plateau method (Sec. A); uncertainty is the standard deviation of $\hat{\alpha}(k)$ over the identified plateau.; RATIO: energy-balance ratio $R_{\text{FDT}} = \langle D \rangle / \langle I \rangle$; equals unity exactly at stationarity; $\hat{\beta}_T$: total MSD exponent; $\hat{\beta}_S$: structural MSD exponent; L_2 : ℓ^2 deviation from Marchenko–Pastur.

ϕ	$\hat{\alpha}$	RATIO	$\hat{\beta}_T$	$\hat{\beta}_S$	L_2
0.0	2.576 ± 0.351	1.000 ± 0.001	1.042 ± 0.006	1.044 ± 0.007	85.54
0.2	2.503 ± 0.325	1.000 ± 0.001	0.989 ± 0.005	0.994 ± 0.005	46.70
0.4	2.393 ± 0.277	1.000 ± 0.001	1.003 ± 0.005	1.016 ± 0.005	23.19
0.6	2.196 ± 0.209	1.000 ± 0.001	1.075 ± 0.004	1.053 ± 0.006	10.36
0.8	2.001 ± 0.151	1.000 ± 0.001	1.028 ± 0.004	1.020 ± 0.004	4.76
1.0	1.868 ± 0.113	1.000 ± 0.001	1.026 ± 0.004	1.020 ± 0.005	3.88

D Results for Financial Assets

Table 3 reports $\hat{\alpha}$ for the AR(1) anchors and for a representative selection of B3 assets spanning five instrument classes. For each real series we also report the shuffle control: the time series is randomly permuted, destroying all temporal correlations while preserving the marginal distribution, and $\hat{\alpha}$ is re-estimated. The systematic shift toward $\alpha_0 \approx 2.58$ in every shuffled series confirms that the measured $\hat{\alpha}$ captures genuine temporal structure and is not an artefact of the marginal return distribution or of the spectral estimation pipeline.

All liquid financial assets yield $\hat{\alpha} \in [1.87, 2.45]$, placing them within the band bounded by the GBM floor and approaching but not reaching the white-noise ceiling. This is consistent with markets that are predominantly efficient at the timescales analysed, yet retain measurable spectral memory not captured by the GBM null model.

Series	Class	$\hat{\alpha}$ (M5)	$\hat{\alpha}$ (H1)	$\hat{\alpha}_{\text{shuffle}}$	$\Delta\alpha$
White noise ($\phi = 0$)	Synthetic	2.576 ± 0.351	—	—	—
GBM ($\phi = 1$)	Synthetic	1.868 ± 0.113	—	—	—
PETR4	Equity	2.023 ± 0.181	1.849 ± 0.122	TBD	TBD
ITUB4	Equity	2.013 ± 0.170	1.973 ± 0.153	TBD	TBD
VALE3	Equity	2.247 ± 0.247	2.245 ± 0.281	TBD	TBD
BBAS3	Equity	1.999 ± 0.164	1.904 ± 0.139	TBD	TBD
ABEV3	Equity	2.445 ± 0.317	2.111 ± 0.205	TBD	TBD
BBDC4	Equity	2.452 ± 0.320	2.333 ± 0.296	TBD	TBD
B3SA3	Equity	2.332 ± 0.278	2.367 ± 0.297	TBD	TBD
WIN\$N	Index future	2.254 ± 0.262	2.312 ± 0.268	TBD	TBD
WDO\$N	FX future	2.240 ± 0.225	2.184 ± 0.234	TBD	TBD
DI1\$N	Rate future	2.116 ± 0.197	1.982 ± 0.160	TBD	TBD
BIT\$D	Crypto future	1.939 ± 0.169	1.816 ± 0.121	TBD	TBD
GOLD11	ETF (commodity)	2.290 ± 0.270	2.099 ± 0.196	TBD	TBD
NASD11	ETF (index)	2.055 ± 0.175	2.072 ± 0.179	TBD	TBD
MXRF11	Real State	3.690 ± 0.907	$7.536 \pm \text{n.a.}$	TBD	TBD
BGI\$D	Commodity future	2.251 ± 0.242	2.327 ± 0.285	TBD	TBD

Table 3: Crossing-time exponent $\hat{\alpha}$ for synthetic reference processes and for B3 financial assets. M5 and H1 columns report independent estimates at the two sampling timescales. The shuffle control destroys temporal correlations while preserving the marginal distribution; the systematic upward shift $\Delta\alpha > 0$ confirms that $\hat{\alpha}$ measures genuine dynamical structure. MXRF11 (real state) and BGI\$D (low-liquidity commodity future) are included for comparison; the anomalously large $\hat{\alpha}$ for MXRF11 at H1 reflects insufficient crossing events at that timescale. Uncertainties are the standard deviations of $\hat{\alpha}(k)$ over the plateau identified by the Hill estimator (Sec. A).

Figure 1 plots the measured AR(1) curve together with the empirical $\hat{\alpha}$ values for a represen-

tative subset of six assets, positioned at their effective AR(1) parameter ϕ_{eff} , obtained by linear interpolation of the measured curve at the asset's M5 $\hat{\alpha}$ value. The resulting ϕ_{eff} values (Table 4) span the range $[0.51, 0.89]$, with commodity and index assets clustering near $\phi_{\text{eff}} \approx 0.5$ – 0.6 and equity and crypto assets near $\phi_{\text{eff}} \approx 0.8$ – 0.9 .

Table 4: Effective AR(1) parameter ϕ_{eff} for selected assets, obtained by linear interpolation of the measured AR(1) curve at the asset's M5 $\hat{\alpha}$ value.

Asset	Class	$\hat{\alpha}$ (M5)	ϕ_{eff}
GOLD11	ETF (commodity)	2.290 ± 0.270	0.51
WIN\$N	Index future	2.254 ± 0.262	0.54
WDO\$N	FX future	2.240 ± 0.225	0.56
DI1\$N	Rate future	2.116 ± 0.197	0.68
PETR4	Equity	2.023 ± 0.181	0.78
BIT\$D	Crypto future	1.939 ± 0.169	0.89

VI Window-Induced Correlations and Finite-Sample Distortions in MSD Estimation

The construction of the Hankel matrix $\mathbf{H}(\tau)$ over a sliding window of size W induces a non-trivial dependence structure in the sequence of operators $\mathbf{K}(\tau)$. In particular, successive matrices share $W - 1$ observations, implying that $\mathbf{K}(\tau)$ and $\mathbf{K}(\tau + \ell)$ are statistically dependent for all lags $\ell < W$. This dependence propagates to the cumulative process

$$\mathbf{X}(\tau) = \int_0^\tau \mathbf{K}(s) ds,$$

and, crucially, to its increments.

Consider the increment over a lag $\Delta\tau$,

$$\delta\mathbf{X}_t(\Delta\tau) = \mathbf{X}(t + \Delta\tau) - \mathbf{X}(t) = \sum_{s=t}^{t+\Delta\tau-1} \mathbf{K}(s).$$

The empirical mean-square displacement (MSD) estimator is defined as

$$\hat{\Sigma}^2(\Delta\tau) = \frac{1}{N - \Delta\tau} \sum_{t=1}^{N-\Delta\tau} \|\delta\mathbf{X}_t(\Delta\tau)\|_F^2.$$

Assuming stationarity, its expectation reads

$$\mathbb{E}[\hat{\Sigma}^2(\Delta\tau)] = \mathbb{E}[\|\delta\mathbf{X}_t(\Delta\tau)\|_F^2],$$

so that the estimator is unbiased with respect to the true MSD. Expanding the squared norm yields

$$\Sigma^2(\Delta\tau) = \mathbb{E}[\|\delta\mathbf{X}_t(\Delta\tau)\|_F^2] = \sum_{i,j=0}^{\Delta\tau-1} \mathbb{E}[\langle \mathbf{K}(t+i), \mathbf{K}(t+j) \rangle_F].$$

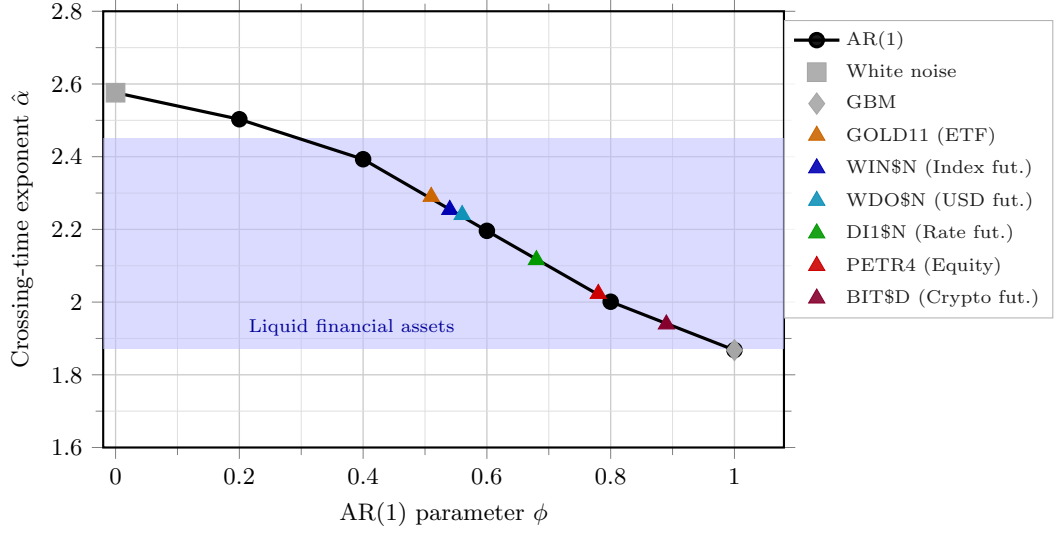


Figure 1: Crossing-time exponent $\hat{\alpha}$ as a function of the AR(1) parameter ϕ . The curve interpolates monotonically between the white-noise ceiling ($\alpha_0 = 2.576$, $\phi = 0$) and the GBM floor ($\alpha_{\text{GBM}} = 1.868$, $\phi = 1$). The blue shaded band $[1.87, 2.45]$ encompasses all liquid financial assets studied (coloured markers, positioned at their effective ϕ_{eff} obtained by linear interpolation at the M5 $\hat{\alpha}$ value; Table 4). Assets span $\phi_{\text{eff}} \in [0.51, 0.89]$, clustering closer to the GBM limit than to white noise, with commodity and index instruments at lower ϕ_{eff} than equities and crypto-linked futures.

Introducing the autocovariance function

$$C_K(h) = \mathbb{E}[\langle \mathbf{K}(t+h), \mathbf{K}(t) \rangle_F],$$

one obtains the standard representation

$$\Sigma^2(\Delta\tau) = \sum_{h=-(\Delta\tau-1)}^{\Delta\tau-1} (\Delta\tau - |h|) C_K(h),$$

which explicitly shows that the MSD depends on the full temporal correlation structure of $\mathbf{K}$.

The windowed construction implies that $C_K(h)$ remains non-negligible up to lags of order W , defining an effective correlation horizon. As a consequence, the MSD exhibits two distinct regimes:

$$\begin{cases} \Delta\tau \lesssim W & \text{correlation-dominated regime,} \\ \Delta\tau \gg W & \text{asymptotic regime.} \end{cases}$$

At short lags, the triangular weighting $(\Delta\tau - |h|)$ strongly couples the MSD to short-range correlations in $\mathbf{K}$, leading to deviations from the asymptotic scaling law. These deviations are not due to a bias of the estimator itself, but rather reflect the intrinsic correlation structure induced by the sliding window.

In addition, correlations between overlapping increments,

$$\text{Cov}(\delta\mathbf{X}_t(\Delta\tau), \delta\mathbf{X}_{t+\ell}(\Delta\tau)) \neq 0 \quad \text{for } \ell < W,$$

increase the variance of the estimator and induce finite-sample distortions in the inferred scaling behaviour. In particular, positive correlations tend to inflate the apparent growth of the MSD at short lags, while negative correlations may produce an apparent subdiffusive regime. These effects vanish asymptotically as $\Delta\tau \gg W$, where both $C_K(h)$ and inter-increment correlations decay.

The magnitude of these finite-sample distortions is therefore controlled by the window size W , which sets the scale of induced dependence. In the main text, the choice $W = 5L$ follows an optimization procedure in which the structural exponent is empirically stable for multipliers in the range $\{4L, 5L, 6L\}$ at the optimal embedding dimension L_{opt} . This choice effectively fixes the correlation horizon introduced by the construction.

The estimation procedure explicitly mitigates these effects in two ways. First, the base lag is chosen as $\Delta\tau_{\text{opt}}$, sufficiently larger than the elementary increment scale. Second, the diffusion exponent is extracted from a log-log regression restricted to the upper 70% of lags, ensuring that the fit is performed within the asymptotic regime $\Delta\tau \gg W$, where correlation-induced distortions are negligible.

Under these conditions, the estimated exponent reflects the intrinsic long-time dynamics of the process, rather than artefacts arising from the sliding-window construction.

Proposed figures for supplementary FigureS1. MSD of (log-log) for all 14 assets, panels arranged by asset class. M5 and H1 shown on same axes. Reference lines and included. FigureS2. vs for all 14 assets at M5 and H1. Vertical band at . FigureS3. vs for all 14 assets. FigureS4. vs (as Fig.3 of main text) for all 14 assets. Vertical band at . FigureS5. MP bulk histogram (as Fig.1 of main text) for all 14 assets. Arranged in 5 panels by asset class. Proposed tables for supplementary TableS1. Complete table of exponents for all 13 assets: , , , , and for both M5 and H1. TableS2. Bulk error relative to MP law for all 13 assets and both timescales. Robustness Checks Sensitivity to . Figure3 of the main text shows results for . All three ratios yield , confirming that does not depend on (i.e., on the number of delay vectors). The value of in the

convergence region varies by across , within the statistical uncertainty. Sensitivity to regression window. Varying the fraction of lags included in the regression from 50Stationarity. We verified stationarity of over the 5-year observation window by computing the Augmented DickeyFuller test on the time series of (total spectral mass), (structural dimension), and (spectral gap). All three quantities are stationary at the 1Non-Gaussianity of bulk projections. Although Eq.(11) of the main text predicts Gaussian projections onto bulk modes, the actual distributions exhibit excess kurtosis of , consistent with the Poisson jump term. This kurtosis is absorbed into the error of the bulk histogram relative to the MP law.

VII Relation to Existing Anomalous Diffusion Literature

The generalized Langevin equation (GLE) framework employed here constitutes a direct matrix-valued extension of the scalar GLE widely used in the study of anomalous diffusion in complex systems. The key structural difference lies in the block organisation of the memory kernel $\mathbf{\Gamma}$, which enables the coexistence of distinct dynamical regimes within orthogonal subspaces of the same operator process. In this sense, normal and anomalous diffusion are not alternative descriptions, but simultaneous projections of a higher-dimensional dynamics.

The connection with linear response theory is explicit. In analogy with the Kubo formalism, the effective friction in the structural subspace,

$$\gamma_s^{\text{eff}} = \hat{\Gamma}_s(0),$$

corresponds to the zero-frequency component of the memory kernel. In the present construction, this quantity is estimated as

$$\gamma_s(\tau) = \frac{1}{L} \text{Tr}(\mathbf{\Gamma}_s(\tau)),$$

providing an operational definition of dissipation directly from the data.

Within this framework, the squared Frobenius norm

$$\|\mathbf{K}\|_F^2 = \text{Tr}(\mathbf{K}^\top \mathbf{K})$$

plays the role of an effective temperature, setting the amplitude of fluctuations through the instantaneous spectral energy of the operator. Although $\mathbf{K}$ is not a velocity in a strict mechanical sense, it acts as the generator of increments of the integrated process $\mathbf{X}(\tau)$, establishing a direct analogy with velocity-driven diffusion. In this interpretation, $\|\mathbf{K}\|_F^2$ is the natural counterpart of $\langle v^2 \rangle$ in classical diffusion theory.

This analogy leads to a matrix-valued Einstein relation for the structural subspace,

$$D_s = \frac{\|\mathbf{K}_s\|_F^2}{\gamma_s^{\text{eff}}},$$

which expresses diffusion as the ratio between fluctuation intensity and effective dissipation. Importantly, both quantities are endogenously determined by the data through the same operator construction, in contrast with standard approaches where noise and memory kernels are specified a priori.

From this perspective, the scalar diffusion coefficient emerges as a reduced description of an underlying operator-valued transport process. The present framework therefore provides a natural bridge between matrix GLE formulations and classical anomalous diffusion theory, while extending the latter to settings where multiple dynamical regimes coexist within a single observable structure.

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